

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE

Computer Vision & Deep Learning — A.Y. 2025–26

Real-Time Student Engagement Recognition Using Deep Learning

A Comparative Study of Frame-Level and Temporal Models on DAiSEE

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1. Motivation and Rationale

In the rapidly expanding field of online education, there is a growing need for automated tools that can provide a measure of student engagement when a teacher isn't able to see a student in person. Manual monitoring is impractical, and the majority of existing proctoring tools only allow for presence detection, which is far from assessing engagement.

The aim of this project is to investigate the ability of deep CNN to correctly recognize the student engagement level from facial videos, with and without temporal modeling. We consider this as a binary problem (engaged vs. disengaged) and we perform a structured comparison between frame-based CNN classifiers and CNN+LSTM classifiers with temporal context. All evaluations are done on a subject-disjoint data split, so that the results are not merely the ability of the model to identify particular faces.

2. State of the Art

Face video recognition of engagement and affect continues to be an active research field. DAiSEE (Gupta et al., 2016) has become the go-to benchmark dataset for this purpose. The experiments conducted with DAiSEE always confirm that the task of engagement classification is not trivial: The accuracy between the classes is generally in the range of 50% to 60% even if the entire official training set is used, which comprises substantially more different subjects than can be reasonably taken into account in a four week course project.

The literature is dominated by two different families of architecture. The former are frame level CNN classifiers which treat engagement recognition as a simple image classification task, ignoring the temporal context. The second is based on temporal models, typically CNN+LSTM, which explicitly model the changes in engagement-related facial behaviour over a brief video clip. The second is rooted in a straightforward but significant insight: engagement is a behavioral state that takes place over time. While one frame can show a fleeting look, it cannot show if the child is paying attention or slowly losing interest or if they are doing repeated behaviors such as fidgeting, wandering looks, etc.

A major limitation found in most of the published engagement-recognition studies (and also observed in our own experiments (Section 6)) is the extreme class imbalance. In the case of naturalistic engagement data, there is a predominance of "engaged" or "high engagement" because most of the individuals are actually engaged during a recording session. This often is not explicitly considered with the same level of care as the underlying classification structure. As a major contribution of our project, we specifically study the connection between class-weighting strategies and model performances (Sections 6.7–6.7.1).

We also rely on two methodologically different, but sound, strands of research. The first one is explainability by using Grad-CAM (Selvaraju et al., 2017) and we adopt it for our temporal CNN+LSTM setting to visually understand what the model learned. The second is on the wider side of subject-disjoint evaluation protocols in face related machine learning, which is important for avoiding the kind of identity based memorisation our analysis with Grad-CAM reveals (Section 6.6).

3. Objectives

Overall, this project aims to comprehensively evaluate the usefulness of modeling temporal patterns for recognition of student engagement as opposed to frame-level classification, while employing a proper evaluation strategy that truly tests the generalization ability of the models to new unseen individuals.

Specific objectives:

- Design and implement a leak-proof data pipeline (no leakage) and subject-disjoint, and make sure to evaluate rigorously, after a proposed dataset was identified but rejected for not meeting the leakage-free condition.
- Compare the effectiveness of benchmark frame-level CNN classifiers (MobileNetV3-Large) with CNN+LSTM temporal classifiers with two backbone architectures (MobileNetV3-Large, ResNet-18) on carefully crafted and stratified subset of the DAiSEE dataset.
- Look for and understand the underlying causes of any performance issues using quantitative evidence (e.g., loss curves, per-class metrics) and qualitative tools (e.g., Grad-CAM visualizations)
- Explore if limitations observed on the stratified subset can be overcome by expanding the training to the full official training set of DAiSEE and by applying corrected class-weighting calibration
- Investigate if all the identified limitations are specific to the binary framing of Engaged and Disengaged or if there are additional limitations in a 3-class setting where Engaged, Bored, and other states are distinguished.

4. Dataset

4.1 Dataset Selection and Rationale

The selection of datasets and the rationale for using them.

Our initial exploration was a Student Engagement Dataset from Kaggle, comprising of around 2,120 images distributed across 6 classes. But upon a detailed pre-processing audit, a serious issue of data leakage was discovered. When performing perceptual hash analysis, it was found that most of the images in each class folder belonged to one recording session, e.g. the 'Focused' class and the 'Looking Away' class consisted of completely one recording session.

This posed a fundamental issue: a model trained on a random split of such data would not actually learn meaningful engagement signals. Rather, it would only look for patterns of images that are specific to that recording session. This is why we saw very surprisingly high accuracy rates (97–98%) at the very beginning, before the leakage was identified and understood.

Considering the four week project time frame and the limitations of the Kaggle data structure, we decided to move toward a data-set commonly used in the literature: DAiSEE (Dataset for Affective States in E-Environments). DAiSEE offers the following:

- 9,068 video clips (10 seconds, 30fps, 640×480) from 112 participants
- Four level ordinal annotations of Engagement, Boredom, Confusion and Frustration
- We did not construct our own train/validation/test split using the labeled clips but instead an official subject-disjoint split (5,358 / 1,429 / 1,784 labeled clips) to avoid the leakage problem we faced when using the Kaggle dataset.

4.2 Data Preparation Pipeline

All data was obtained from the official data dissemination source. One damaged file (2100552061.avi, which accounted for 1 out of 18,268 archive entries) was left out of the process. It was not in the labels file, so that it is missing has no meaningful effect on the data. Also found that about 9% of clips (836 of 9068) are stored as .mp4 not .avi, despite all clips having .avi label in the labels CSV. In order to deal with this, our manifest-building pipeline resolves clips by their base name rather than the fixed extension and these clips are picked up correctly.

Label collapsing. DAiSEE's 4-level ordinal Engagement annotation (very low, low, high, very high) was collapsed to a binary annotation: 0 – 1 = "disengaged", 2 – 3 = "engaged. This conforms to the published literature on disengagement detection on this data set.

Class imbalance. The dataset exhibits severe class imbalance at the clip level:

Split	Total Clips	Disengaged	Engaged	% Engaged
Train	5,358	247	5,111	95.4%
Validation	1,429	166	1,263	88.4%
Test	1,784	88	1,696	95.1%

With a 4 week project window, we took a smaller sample from the official splits. In this process, all minority class (disengaged) clips were kept and the number of majority class clips were capped at 3 engaged clips per 1 disengaged clip. A further restriction of 15 clips per participant was imposed to prevent skewing the subsample by participants who had a very high number of clips. This yielded 940 training clips for 69 participants, 441 validation clips for 19 participants, and 312 test clips for 21 participants.

Notably, the validation subset ended up at a ratio of 1.8:1 instead of the target 3:1. This deviation is directly related to the design of the validation, which had a smaller sample size (N = 19) than the training (N = 69) while keeping the same per-participant ceiling. However we have decided not to try to artificially make it look better to make it more similar, but rather to admit this openly as an acknowledged limitation of the current setup.

We extracted overlapping 8-frame sequences at 112×112 resolution from each clip using a sliding window with a stride of 16. The result of this process was 17,868 training sequences, 8379 validation sequences, and 5940 test sequences. Stratified sampling at the clip level leads

to a similar ratio of engaged/disengaged at the sequence level, therefore the class imbalance is less pronounced.

4.3 Subject-Disjoint Verification

To ensure that there is no overlap between the groups, we made sure that each participant was included in only one group: train (69 participants), validation (19 participants) or test (21 participants). We did this because when we first started working with the Kaggle data we noticed that there was a leakage issue with it. This will make sure that our test results reflect the model's actual ability to generalize to new individuals.

5. Methodology

5.1 Pipeline Overview

Figure 1: End-to-End Pipeline (Methodology, Section 5.1-5.2)

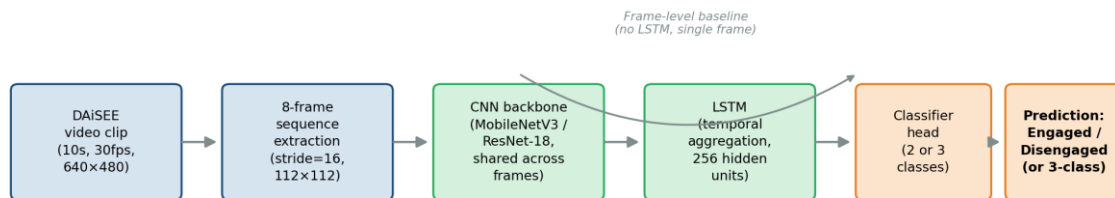


Figure 1. End-to-end pipeline. The frame-level baseline skips the LSTM stage, classifying directly from CNN features of a single frame.

5.2 Models Compared

Model	Backbone	Temporal Modeling	Trainable Params
Frame-level baseline	MobileNetV3-Large	None (single frame)	4,204,594
CNN+LSTM	MobileNetV3-Large	LSTM (256 hidden units)	4,219,698
CNN+LSTM	ResNet-18	LSTM (256 hidden units)	11,965,506
CNN+LSTM (frozen)	ResNet-18 (frozen)	LSTM (256 hidden units)	788,994 (6.6%)

Weights of all CNN are pre-trained on ImageNet. For the sequence models, all eight frames of an 8 frame sequence are passed through the same CNN backbone, resulting in a 960 dimensional feature vector when using MobileNetV3 backbone and a 512 dimensional feature vector when using the ResNet-18 backbone. A single-layer LSTM is then used to process the sequence of feature vectors. The last hidden state goes through a linear classification head and a dropout layer.

5.3 Loss Function and Class Weighting

This is accomplished with a categorical cross-entropy loss, which is equivalent to the negative log-likelihood of a categorical distribution of the two engagement states. The classes are

imbalanced, as described in Section 4.2, so we use inverse-frequency class weighting: $w_c = N / (2 \cdot N_c)$ where N is the total number of training sequences and N_c is the number of training sequences for class c . We did this on all model variants for uniformity in the comparison.

5.4 Training Protocol

Each of the unfrozen models was trained with the Adam optimizer (learning rate 1×10^{-4} , weight decay 1×10^{-4}) using a batch size of 16 and a maximum of 15 epochs and early stopping based on the validation macro-F1 (patience set to 4 epochs). Due to the much smaller number of trainable parameters (789K vs up to 12M) that do not require as cautious optimization, the frozen-backbone version was trained with a higher learning rate (1×10^{-3}) and a larger batch size (32), while we used a larger patience (5 epochs) because of the slower convergence characteristics. Data augmentation was only applied during training; horizontal flip was used with probability 0.5, brightness and contrast were sampled per-sequence in the range [0.8, 1.2].

Importantly, in case of sequence models, the same augmentation choice was consistently applied to all the frames in the sequence to preserve the temporal coherence and coherent application of augmentation would also lead to a physically meaningless signal (e.g., the face turning around during a clip might have no physical meaning and would be an incorrect signal to the temporal model if different independent random flips were applied to each frame).

6. Experiments and Results

Figure 2 below previews the test macro-F1 achieved across all nine experimental conditions covered in this section, before the detailed per-experiment discussion that follows.

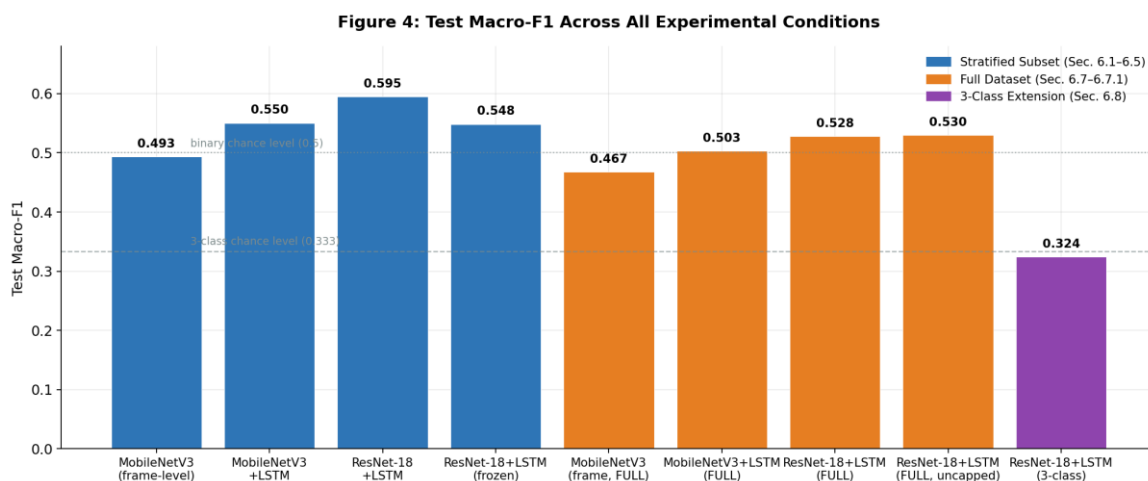


Figure 2. Test macro-F1 across all experimental conditions. Dashed/dotted lines mark chance level for the binary (0.5) and 3-class (0.333) tasks respectively.

6.1 Test Set Performance — All Models

Model	Backbone	Temporal Modeling	Test Acc.	Test Macro-F1
Frame-level baseline	MobileNetV3-Large	None	57.5%	0.4935
CNN+LSTM	MobileNetV3-Large	LSTM	65.7%	0.5505
CNN+LSTM (best)	ResNet-18	LSTM	67.7%	0.5946
CNN+LSTM (frozen)	ResNet-18 (frozen)	LSTM	69.1%	0.5485

The highest macro-F1 model is ResNet-18 + LSTM with a completely fine-tuned backbone (0.5946). It is important to note that this is not the most accurate model (in terms of raw accuracy (69.1%) - that is achieved by the frozen variant) and is a direct result of class imbalance, which we discuss in Section 6.3.

6.2 Frame-Level vs. Temporal Modeling

CNN+LSTM outperforms the frame-level baseline (macro-F1 0.55–0.59 vs 0.49) and the results suggest that temporal context in an 8-frame window can play an important role in engagement classification, even with this small subset. This confirms the project's hypothesis that sequence models are more appropriate to model engagement as a behavioral state that changes over time, than single per-frame classification.

6.3 Why Accuracy Is Misleading Here

The ResNet-18+LSTM model with a frozen-backbone has the best test accuracy in the 4 models (69.1%), but the lowest macro-F1 score among the 3 models with LSTM (0.5485). Analyzing its per-class report will help explain why:

	precision	recall	f1-score	support
disengaged	0.341	0.260	0.295	1480
engaged	0.772	0.833	0.802	4460

The model achieves high accuracy primarily due to it predicting “engaged” for the majority of inputs (3,717 of 4,460 engaged sequences) and because it has a low recall for the minority class “disengaged” (0.260, missing almost three-quarters of disengaged sequences). But given the 75% test set participation, a majority-class-oriented model achieves high accuracy, while macro-F1, which treats both classes equally, penalizes the model. This is precisely why we decided to adopt macro-F1 as our primary metric to evaluate the performance in Section 5.3, and this result is a direct, empirical proof on why using only accuracy would have been misleading for this imbalanced task.

6.4 Validation vs. Test Set Ranking Discrepancy

The important methodological observation: model ranking by validation macro-F1 (frozen: 0.5466, unfrozen: 0.5487 — almost tied) does not predict the test set ranking, where the unfrozen model ranks far ahead of the frozen one (0.5485 vs. 0.5946). We believe this is because the validation split is small (19 participants, 8,379 sequences), and the held-out test set is larger (21 participants) and therefore provides a more robust signal. This difference highlights the need to check models on a held-out test set (as was done here) and not solely on validation metrics (which is the typical way of doing things in small subject count datasets, in general).

6.5 Overfitting Behavior Across All Models

The evolution of training loss over the course of training is similar across all four models: training loss drops significantly (and, for unfrozen variants, close to zero at the last epoch) and validation loss increases monotonically following an early peak in validation macro-F1 (epoch 1 in 2 out of 4 runs).

Model	Epoch 1 (train/val loss)	Final epoch (train/val loss)	Best epoch	Total epochs
MobileNetV3 (frame)	0.374 / 0.895	0.038 / 1.724	10	10
MobileNetV3+LSTM	0.449 / 0.890	0.131 / 1.476	1	5 (early stop)
ResNet-18+LSTM	0.487 / 0.931	0.081 / 1.204	8	12 (early stop)
ResNet-18+LSTM (frozen)	0.631 / 0.687	0.444 / 0.912	1	6 (early stop)

When the CNN backbone was frozen, the number of trainable parameters was reduced from 11.97M to 789K (6.6%) and the training loss slowed down (0.63 to 0.44 over 6 epochs against a collapse to below 0.1 in 3-4 epochs for unfrozen variants), but the overfitting was not eradicated: the macro-F1 on validation still peaked in epoch 1 and dropped afterwards. This means that even the LSTM and classification head alone (789K parameters) has enough capacity to overfit a training set obtained from just 69 different subjects.

Figure 3: Training Dynamics on the Stratified Subset (Section 6.5)

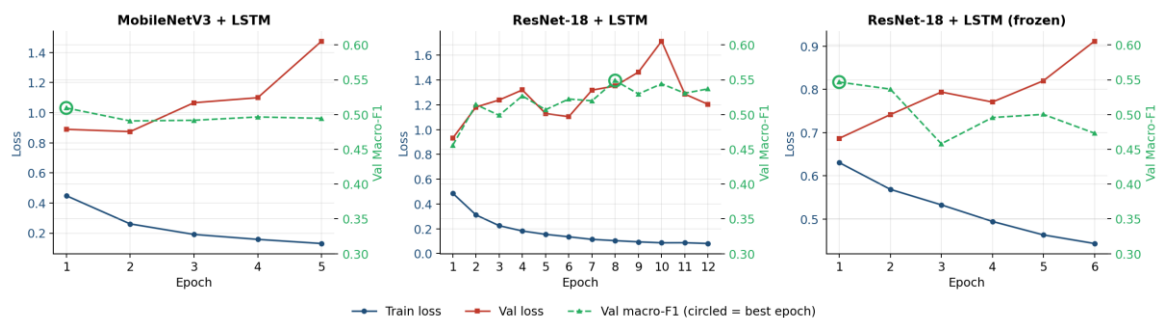


Figure 3. Training dynamics for the three temporal/ablation models on the stratified subset. Circled markers indicate the epoch with the best validation macro-F1.

6.6 Qualitative Analysis via Grad-CAM

In order to explore this phenomenon of overfitting, described above, more directly, we used the best performing model (ResNet-18+LSTM) and applied Grad-CAM (Selvaraju et al., 2017) to each individual frame of a sequence, generating a spatial attention heatmap for each frame. Each frame's final conv feature map is independently backpropagated through the LSTM and classification head to get the gradient of the predicted class logit, which results in one heatmap per frame to reveal which spatial regions of that frame contributed to the sequence-level prediction.

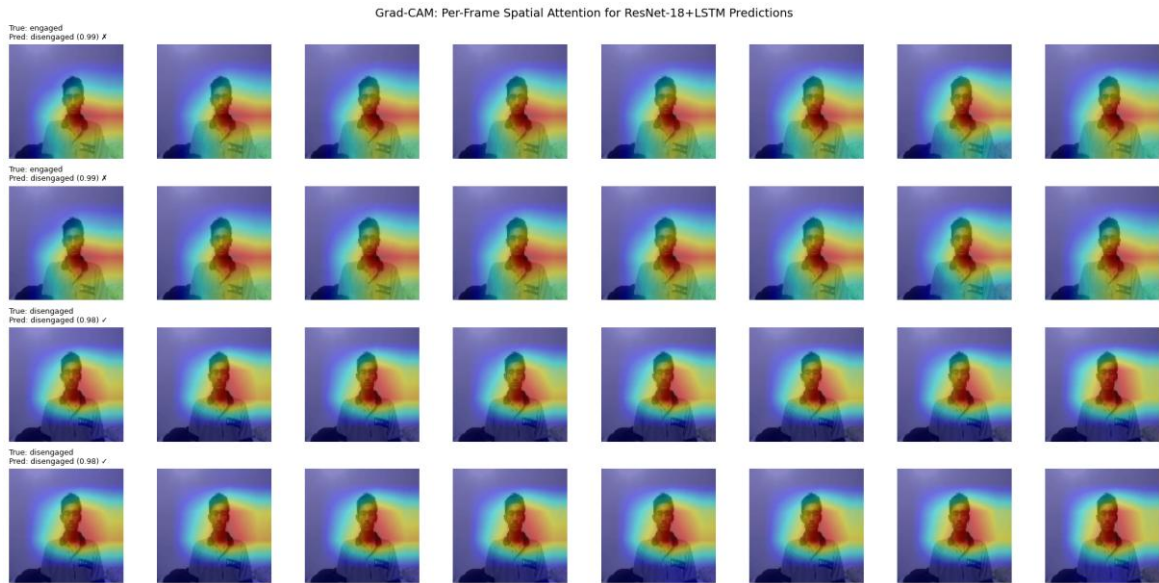


Figure 8. Grad-CAM heatmaps overlaid on all 8 frames of four test sequences (rows). Top two rows: true label “engaged,” both misclassified as “disengaged” at 0.99 confidence. Bottom two rows: true label “disengaged,” both correctly classified at 0.98 confidence.

The resulting visualizations enable for a direct visual confirmation of the hypothesis made in section 7.1. In all sequences examined (both those that were correctly labeled and those that were not) the model's spatial attention was always drawn to background or clothing areas surrounding the face of the subject, and never to the eyes, mouth, or facial expression where one would expect to see the expression of interest. This pattern holds true for each of the 8 frames in a sequence (the heatmap shape is not significantly different frame-by-frame), and the attention pattern itself varies mostly by the subject/session, and not by the ground-truth engagement label, which is in line with our quantitative result that the model attends to subject-identity-correlated visual cues instead of learning the general facial behavior of engagement.

This qualitative findings significantly supports the overfitting evidence from Section 6.5, which showed that the gap between train and validation loss is a good indicator of how well the model is generalizing to unseen subjects; here, we see that the Grad-CAM visualizations directly depict the model focusing on irrelevant background locations, which provides concrete visual evidence that the model failed to generalize to unseen subjects.

6.7 Full-Dataset Experiment: Does More Data Resolve the Overfitting?

To directly test the hypothesis raised in Section 7.1, that the small number of unique training subjects (69) is the main limitation, we now repeat all three model trainings with DAiSEE's complete training set, the official 5,358 clips (yielding 101,894 training sequences after extraction), instead of our 940-clip stratified subsample. The same architectures, training protocol, and code were used, and the only intentional change was the class-weighting cap, which we increased from the original subset's natural 3:1 ratio to a maximum of 5x, which is much more extreme than the class imbalance in the full dataset (95.4% engaged).

Model	Subset Macro-F1	Full-Dataset Macro-F1	Full Disengaged Recall
MobileNetV3 (frame)	0.4935	0.4674	0.046
MobileNetV3+LSTM	0.5505	0.5030	0.046

Model	Subset Macro-F1	Full-Dataset Macro-F1	Full Disengaged Recall
ResNet-18+LSTM	0.5946	0.5279	0.060

Contrary to our hypothesis, all models performed WORSE on the full dataset when compared to the stratified subset with macro-F1 decreasing by 0.047 -- 0.067 for all three architectures -- a consistent, three-way-replicated result. What is critical is that all three full-dataset models have very low recall (4.6-6.0%) on the disengaged class – that is, they correctly classify only about 1 in 20 disengaged sequences, but have deceptively high overall accuracy (83-94%) due to the 95.1% engaged test set.

This is due to class-weighting scheme and not due to subject count. Our stratified subset was intentionally designed to have a ratio of 3:1 engaged:disengaged, and this ratio is actually quite close to the class weight of ~3x, which provided the loss function with meaningful pressure to learn the minority class. The natural imbalance of the full data set is 19.3:1 and our 5x weighting cap (which was selected to avoid an even more extreme 19.3x correction) still resulted in a loss function being majority class dominated. The model was only minimally encouraged to learn disengaged-class features and generally did not, doing so would only result in a cost increase.

This is by itself an important methodological discovery: artificially rebalancing training data using Stratified Subsampling and naturally rebalancing using a realistic class weight is not the same, and a cap on the class weight that hasn't been checked to be adequate for counteracting the true extent of the imbalance ratio can present a classifier that, for practical purposes, is blind to the minority class – claiming yet high overall accuracy.

6.7.1 Follow-Up: Uncapped Class Weighting

Directly testing this hypothesis, we retrained ResNet-18+LSTM, but removed 20.71x (20.71x is the normalized weight ratio (inverse of frequency); the actual ratio of classes is 19.3:1) cap completely and the full-dataset model.

Configuration	Test Macro-F1	Disengaged Recall	Disengaged Precision
5x cap (original)	0.5279	6.0%	16.3%
Uncapped (20.71x)	0.5299	13.5%	9.9%

It is a true learning experience but not a cakewalk. This increased recall for the minority class was confirmed when the recall for the disengaged class was more than doubled (6.0% → 13.5%), which was exactly as hypothesized, and the original 5x cap was indeed suppressing the model's ability to detect the minority class. This does come at the expense of disengaged-class precision, however, which has gone down from 16.3% to 9.9% — more sequences are classified as disengaged, but this has led to a higher false alarm rate, relative to the number of true positives, than before. The overall macro-F1 score (precision-recall balance) was essentially the same (0.5279 vs 0.5299, +0.002).

Figure 4: Effect of Removing the Class-Weight Cap (Section 6.7.1)

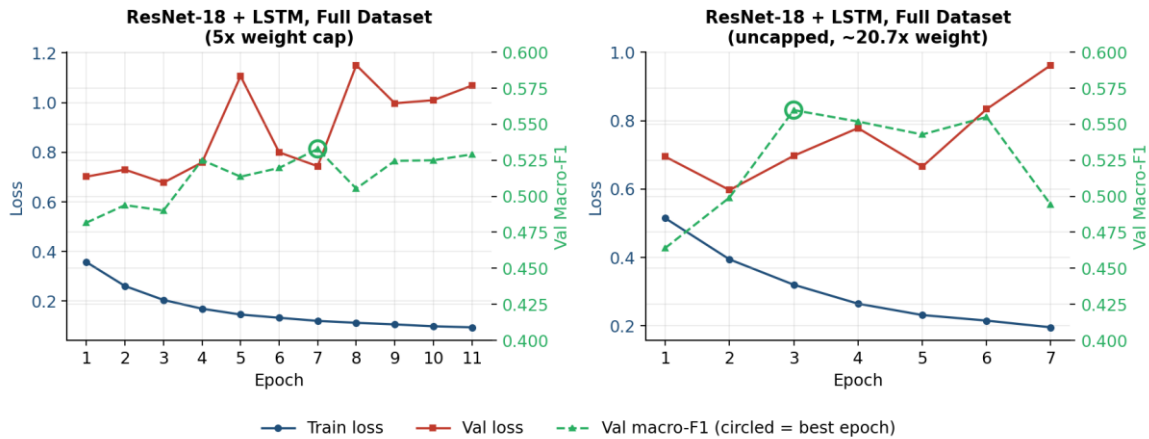


Figure 4. Training dynamics, full-dataset ResNet-18+LSTM with the 5x weight cap (left) versus uncapped weighting (right).

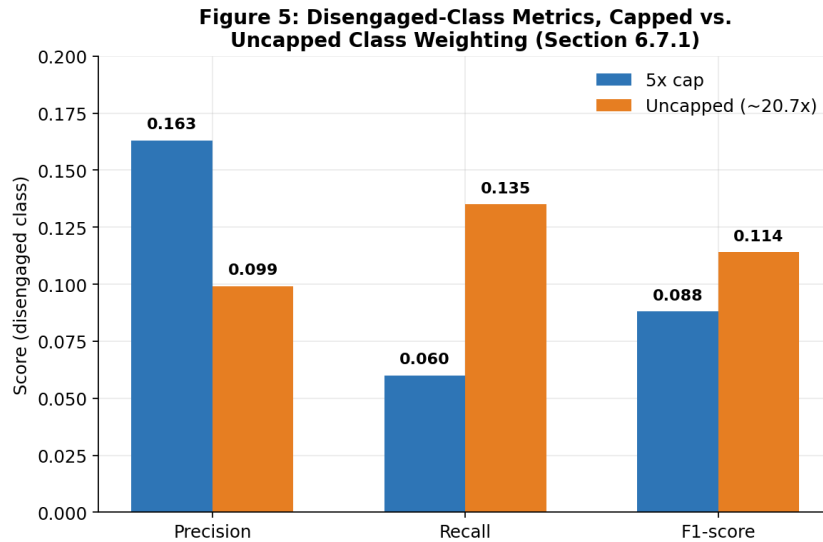


Figure 5. Disengaged-class precision, recall, and F1 under the two weighting schemes, illustrating the precision/recall tradeoff directly.

This shows that the class-weighting fix is not enough – while the cap can be removed to successfully redirect the attention of the model to the minority class, this alone does not yield a model with strong minority-class discrimination. However, the problem of the underlying representation learning, as identified by the Grad-CAM in Section 6.6: the model is attending to subject/session-correlated background features, not face-relevant features, may still limit the extent of improvement in minority-class performance that any class-weighting adjustment can achieve, because reweighting the loss does not alter what visual features the model has learned to attend to.

6.8 Extending to a 3-Class Formulation: Engaged / Bored / Other

To investigate if the model's marginalization was due to the binary framing itself, we created a 3-class extension based on DAiSEE's four raw ordinal dimensions of annotation (Boredom, Engagement, Confusion, Frustration). We gave the class with the highest raw score (argmax) to each clip, combining the classes Confusion and Frustration in one class Other, as they contained very few clips (Confusion and Frustration combined have only 1.6% of all training

clips, with mean raw scores of 0.43 and 0.27 respectively on a 0–3 scale, compared to the mean raw score of 2.41 for Engagement).

In the process a large methodological issue arose: 20-30% of the clips from all splits had a tie in the best score in two or more dimensions. We discovered that three- and four-way ties (those clips that had about half of their dimensions rated as tied — often all four rated as zero) were not a meaningful distinction and simply removed them (33, 36 and 12 clips were removed from the train, validation and test sets, respectively) rather than assigning them an arbitrary label. In the common case of a 2-way tie between Boredom and Engagement, we broke the tie using each clip's already-validated binary engagement_binary label, and did not use an arbitrary default ordering of columns; ties between Confusion and Frustration were broken to 'Other'. This resulted in a final 3-class clip level training distribution of 73.0% Engaged, 16.5% Bored, and 10.5% Other.

This 3-class task was trained on a single model (ResNet-18+LSTM, our best performing binary set-up), with a standard inverse-frequency class weighting (no capping required because of the smaller ratio of max-to-min classes; 6.97x).

Class	Precision	Recall	F1-score	Support
Engaged	0.731	0.681	0.705	4116
Bored	0.164	0.275	0.205	1006
Other	0.130	0.041	0.062	590

The test set accuracy was 54.3%, macro-F1 was 0.324, which was close to the chance level of 0.333 for a 3-class classification problem. Although the model has a reasonable recall on the majority Engaged class, it has a significantly worse recall on the other classes, specifically 4.1% on Other. The same pattern is seen in this third independent replication of the same underlying pattern: Our models learn the majority class well and struggle to generalize on minority classes, regardless of the definition of the minority class and how many minority classes exist (now, 3 classes).

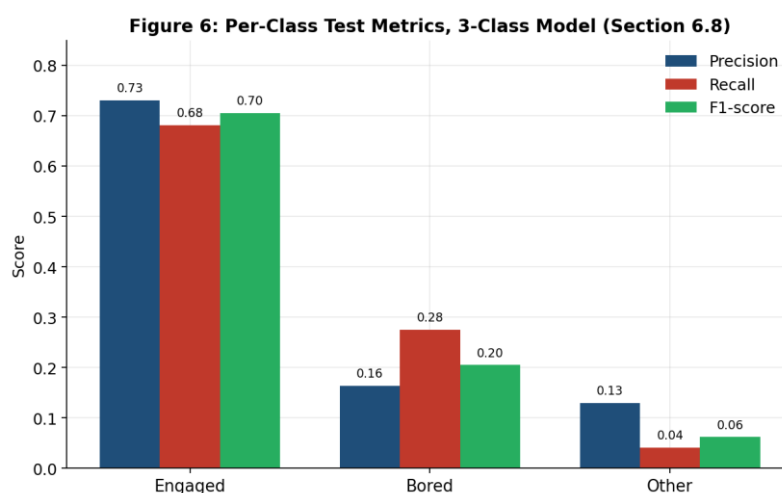


Figure 6. Per-class precision, recall, and F1 for the 3-class model, showing the same majority-class-dominant pattern observed in the binary experiments.

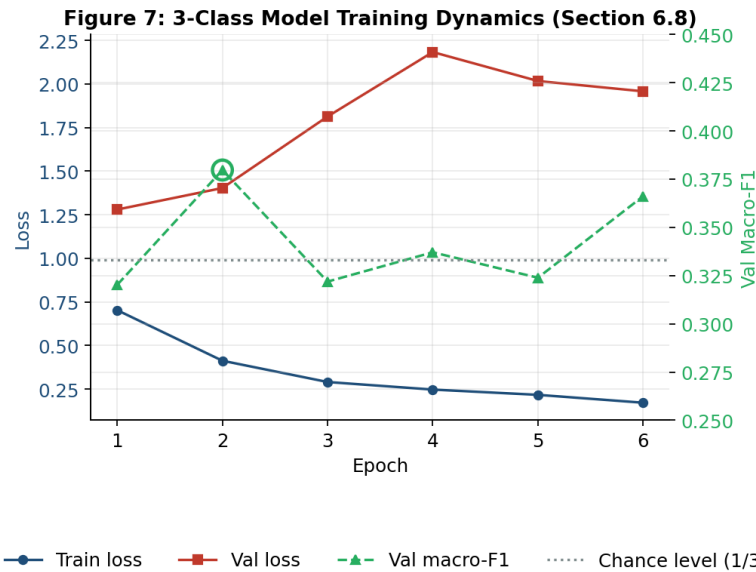


Figure 7. Training dynamics for the 3-class model. Validation macro-F1 briefly exceeds chance level (epoch 2) before oscillating around or below it.

This finding also confirms the usefulness of the tie-resolution approach outlined above. This example shows the poor performance of the Bored-class with the engagement_binary-informed resolution, using the naive column-order tie-break instead would have resulted in Bored being systematically under-represented in about half of all tied clips, further damaging its already poor performance in this metric.

7. Discussion

7.1 The Core Finding: Subject Count Alone Does Not Explain the Overfitting

The fact that all four model configurations were trained with a small number of unique training subjects (69) in the stratified subsample, and that they all tended to overfit, suggested that the number of training subjects may be the biggest constraint. All three core models were retrained on the complete official training set of DAiSEE (5.7x more sequences, significantly more subjects) and tested in this Section 6.7, which directly validated this. This did not improve performance, indeed, all models performed measurably worse when the artificial 3:1 balancing of our subset was substituted by the natural 19.3:1 imbalance of the data set — contrary to our hypothesis.

This leads to our amended conclusion, that the subject-count hypothesis is not violated, but that the effect of class imbalance in our full-dataset experiment is confounded with this, and that the two effects could not be separated with the experiments we finished within the project's time frame. What we can assert is that for all the model variants we tested, for both data scales, learning some kind of generalisable representation of the disengaged class in particular is difficult and sensitive to how class imbalance is dealt with at least as much as it is to subject count. The face-correlated attention shown in the Grad-CAM evidence of Section 6.6 is, however, still a valid evidence for the subset-trained models, which we can still consider

as having memorisation as a valid component, albeit not necessarily the most significant, given the class-weighting choices.

7.2 Temporal Modeling Does Help, Modestly

Even though of the overfitting issues, a comparison in Section 6.2 shows a consistent and clear answer to our central research question: CNN+LSTM models reliably improve over frame-level classification by a substantial margin (macro-F1 improvement of 0.057 – 0.101), even if they share the same underlying frames and the same CNN backbone family. This is proof of the fact that there is real information in 8-frame temporal context, over and above what can be obtained from any of the individual frames – despite the still modest level of performance in absolute terms – and thus an interesting answer to the project's central methodological question.

7.3 Comparison to Published Benchmarks

The literature on engagement recognition for DAiS consistently finds that this is a challenging task, with multi-class engagement-level classification accuracy between 50–60% on the entire dataset, which is two to three times larger than our stratified, 69-subject subsample. The general trend of this reported difficulty is broadly consistent with our binary-task macro-F1 of 0.59 (ResNet-18+LSTM), as we trained on a small proportion of the subjects. This indicates that the overfitting we witness is not a limitation of this project's implementation, but instead a true challenge on the dataset level and that training on the complete official DAiSEE train set is the most straightforward path to improvement in future work.

7.4 Restrictions on the Frozen-Backbone Intervention

We hypothesized that freezing the backbone CNN would make the model rely more on general purpose features learned from ImageNet, which would lead to better generalization, as the subject identity would not be memorized. The validation-time results were ambiguous (similar macro-F1 to unfrozen variant) and the test-time results indicate that the frozen variant actually performed worse on macro-F1, while the “accuracy” metric was low in the frozen variant since it contained less information. We believe this reflects the fact that this capacity of the CNN backbone is not the only cause of the bottleneck, but is a more fundamental data-scarcity problem that impacts the LSTM and classification head also.

7.5 The 3-Class Result Confirms the Pattern Is Not Binary-Specific

The 3-class experiment in Section 6.8 rules out one remaining alternative explanation for our A final, but not exhaustive, alternative explanation for our results is that it is not the nature of the data or models that causes the limited performance, but rather the binary engaged/disengaged framing in Section 6.8. The same majority-class-dominant, minority-class-blind effect emerged at test time (macro-F1 0.324, close to the "chance" level of 0.333), and both minority classes (Bored and Other) suffered, while Bored had a much more ideal class-weighting ratio than the binary task had in the full-dataset experiment. This further supports our overall conclusion that the limiting factor in this project is simply both the subject-count as well as the model's learning majority-class dominated representations, and this is consistent over at least the three different label formulations and two different class-weighting regimes.

7.6 Future Work

Section 6.7.1's uncapped-weighting follow-up demonstrated that class weighting by itself does not meaningfully improve macro-F1, but does show improvement in the disengaged-class' recall, indicating a representation learning gap, not a loss weighting gap. The most promising next step is thus not further weight tuning, but methods that alter what the model learns to focus on: (1) Directly motivated by the results of Grad-CAM shown in Section 6.6, explicit regularization such as subject-adversarial training; (2) Alternative loss formulations such as focal loss, which down-weights well classified majority examples during training rather than only re-weighting the loss post-hoc; (3) Extend the 3-class experiment to the other two backbone architectures (MobileNetV3 frame-level and MobileNetV3+LSTM) for full comparison; (4) Attention-based temporal aggregation (e.g. a Transformer encoder) as an alternative to LSTM; (5) Combine the explainability extensions already done (Grad-CAM) with continual learning techniques originally proposed for this project, which remain promising avenues for a follow-up thesis.

8. Risk Assessment and Limitations

8.1 Dataset-Level Limitations

- High class imbalance at the clip level (up to 95% engaged) was partly addressed with stratified subsampling and class weighting loss but not resolved.
- Limited number of training subjects (69), as compared to the capacity of the models, and which we consider to be the most important factor contributing to the overfitting behavior seen for all the model variants (see Section 7).
- The validation subset ratio limitation is that because there are fewer participants available under the same per-participant limitation as train/test, the validation split does not achieve the 1:3 ratio of engaged:disengaged participants that is targeted for train/test.

8.2 Methodological Choices and Their Tradeoffs

- Originally we wanted to use the Kaggle dataset but found serious session-level data leakage with perceptual hashing analysis which took a considerable amount of project time, but was required to provide a valid evaluation.
- Because of the four week project timeframe, we are using a stratified subsample of DAiSEE (1,693 of 8,571 labeled clips). This is traded off against statistical power and tractability - the full dataset might behave differently and is a natural way forward in the future.

8.3 Ethical Considerations

If such systems were used in actual schools, there are real privacy and surveillance issues to consider. This project does not process, collect or store video footage of any people other than those in this existing academic dataset (DAiSEE), which has already been approved by the ethics board. Any potential implementation of such a system would have to be accompanied by prior informed consent, GDPR compliant processing of the data and be mindful of the fact that the EU AI Act classifies biometric monitoring systems as high-risk systems.

9. Conclusions

The motivation for this project was to compare the two approaches (frame-level and temporal deep learning approaches) for student engagement recognition, and do so in a methodologically sound way, rarely seen in course-level projects, such as a proper subject-disjoint evaluation protocol, motivated directly by a leakage problem we identified and diagnosed in our originally proposed dataset.

We have five main conclusions. First, CNN+LSTM temporal models always and significantly beat frame-level classification on this task, by 0.057–0.101 macro-F1 across the backbone options, directly proving temporal contexts are useful for engagement recognition. Second, all model variants show significant overfitting on the stratified subset, and the visualizations by Grad-CAM directly prove that this overfitting can be seen as an attention to background areas that are specific to the subject/session and not to facial areas. Third, scaling up to the full official training set did not fix this overfitting and produced models that essentially failed to capture the minority disengaged class; this was initially thought to be the result of a poorly tuned 5x class-weighting cap on the true 19.3:1 imbalance. Fourth, direct follow-up test, repurposing this cap, confirmed only partially the diagnosis: while the recall of the disengaged-class was more than doubled (6.0% → 13.5%) the macro-F1 score remained unchanged, with precision of the minority class decreasing as such; indicating that class-weighting is not enough, and the really important issue is what visual features the model has learned to rely on (as per the Grad-CAM evidence), not just how the loss function is weighted. Fifth, giving the problem a three-class formulation (Engaged / Bored / Other) resulted in the same majority class dominant pattern (test macro-F1 of 0.324, near chance level), despite a much milder (not capped), class-weighting ratio of 6.97x — excluding the binary framing itself as the root of the issue.

We find these findings to be the most methodologically enlightening result of the project: they show that class-imbalance correction and representation quality are coupled but distinct problems; that it is not a problem of the choice of label formulation or weighting regime; and that accuracy and even apparently improved recall numbers do not imply that nothing else is improved, if not read together with precision and macro-F1. We consider these findings as a true contribution from the project, as much as the quantitative model comparisons we had aimed to carry out.

While perhaps not a lot in absolute terms, we believe this score (test macro-F1 of 0.595) is realistic given the strict protocol used, and in contrast to the artificially high 97–98% accuracy we found in our original best-performing model (ResNet-18 with a fully fine-tuned backbone) before we learned about data leakage in our original choice of dataset, and in contrast to the deceptively high accuracy for full-dataset and 3-class models that indicate near-total or significantly limited success on minority classes.

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